Physics 1A Discussion (Week 6)

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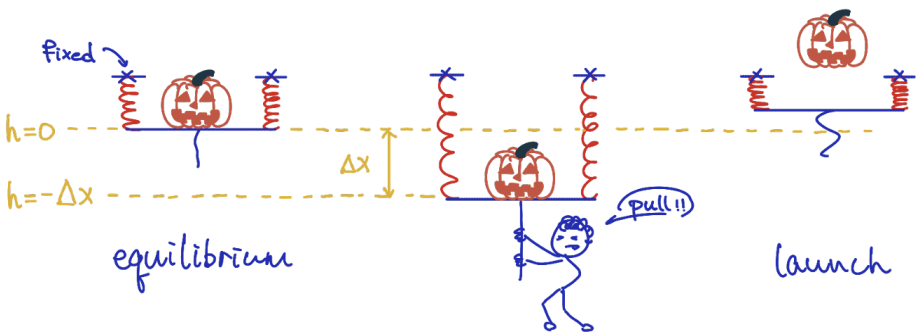
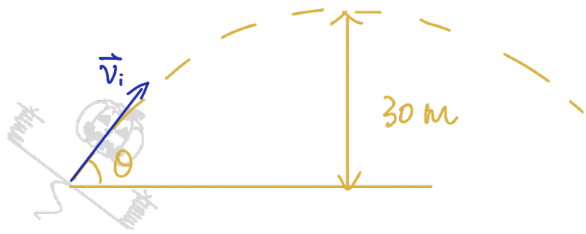
Jinyan invented a launch machine as shown. Two identical springs are attached to a plate and he is trying to launch a pumpkin using this launcher. Jinyan spends 500 N to pull the plate a distance $\Delta x = 2\,\mathrm{m}$ from the equilibrium position of the launcher. Then he releases the plate to launch the 1 kg pumpkin. Neglect friction in this problem.



(a) Find the speed of the pumpkin when the launcher is back to its equilibrium position.

(b) Find the maximum height that the pumpkin can reach.

(c) If the pumpkin is launched at an angle as shown (assuming that it leaves the launcher with the same speed $|\vec{v}_i|$ as computed in part (a)), find the angle θ relative to the ground such that Jinyan can launch the pumpkin to reach a maximum height of 30 m.



Hint: For part (a), first compute the spring constant of the springs. Then apply energy conservation. Part (b) is a straightforward application of energy conservation. Part (c) is a little tricky, Pythagoras said $v^2 = v_x^2 + v_y^2$, and we can use that in our application of energy conservation.